

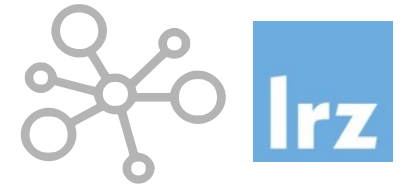


Leibniz Supercomputing Centre
of the Bavarian Academy of Sciences and Humanities

The background of the slide is a photograph of a modern, multi-story building with a glass and metal facade. The building is partially obscured by a dark blue semi-transparent rectangle that contains the title text. The sky is overcast with grey clouds. In the foreground, there are some trees and a street with a few people walking.

Introduction to Julia

General Programming Software Design Principles



SRP

OCF

LSP

ISP

DIP

DRY

KISS

POLA

YAGNI

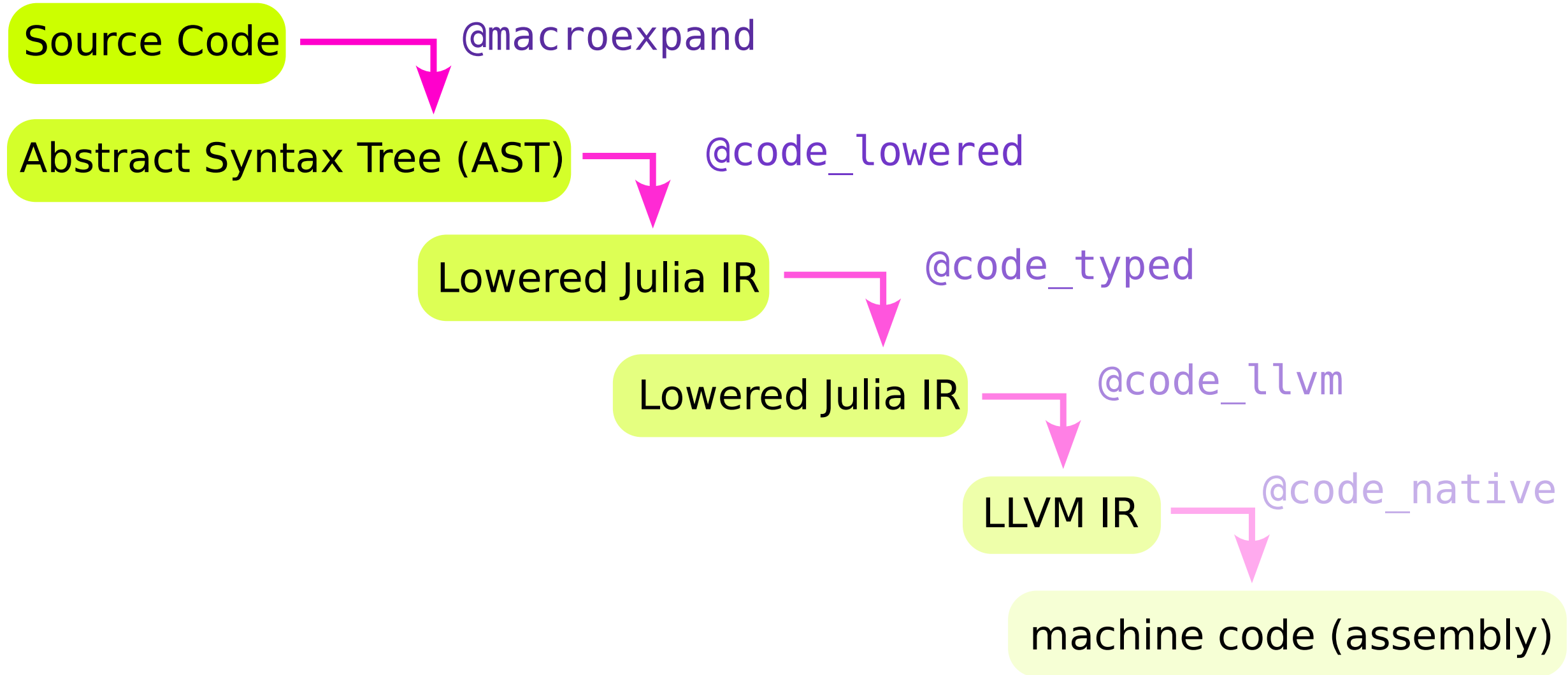
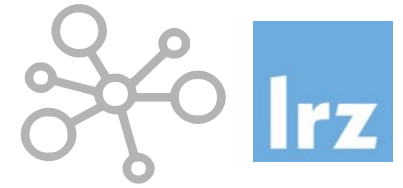
POLP

General Programming Optimization Guidelines



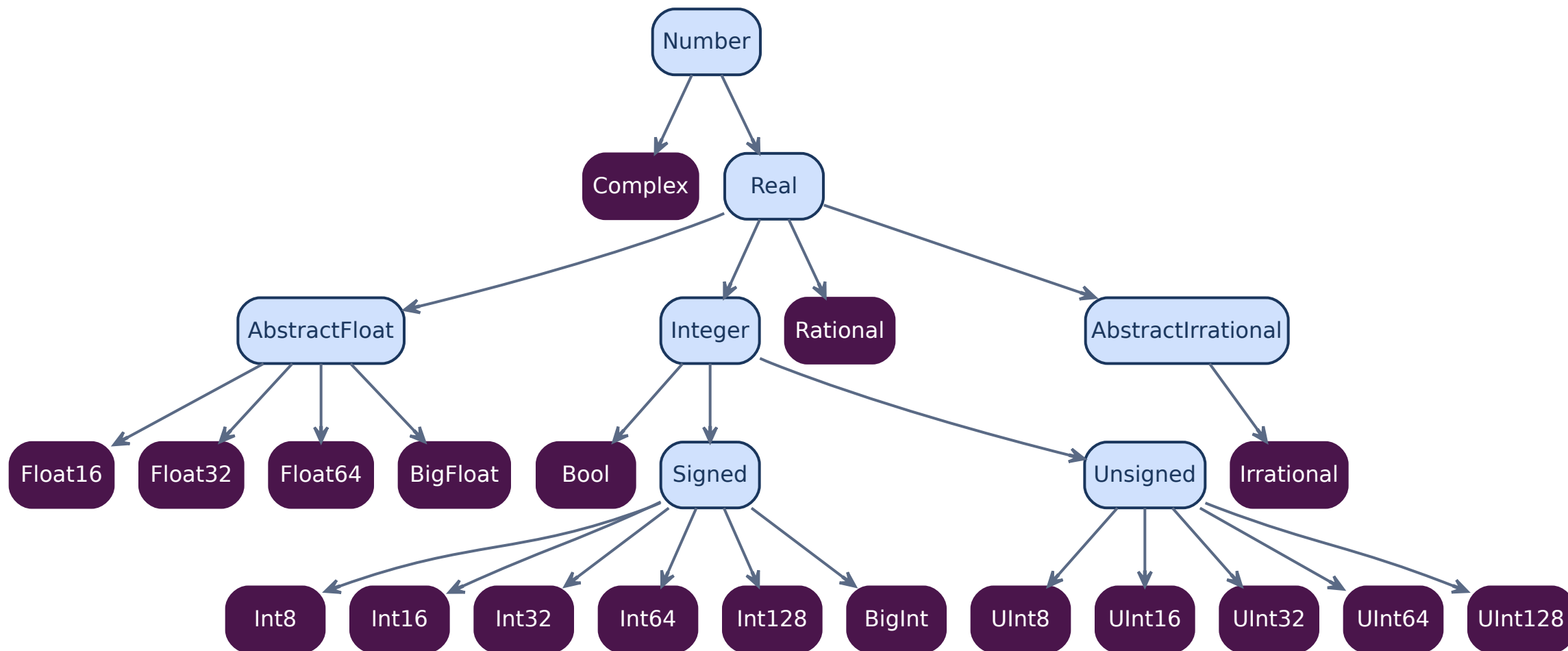
- No premature optimization! Measure!
 - Optimize only critical regions! (20/80 rule)
 - First CORRECT, second FAST!
 - Write communicating, maintainable code!
 - Refactor often! Understand your code!
 - Use programming patterns if applicable!
- Adhere to the Software Design Principles!

Julia Compilation Pipeline Stages



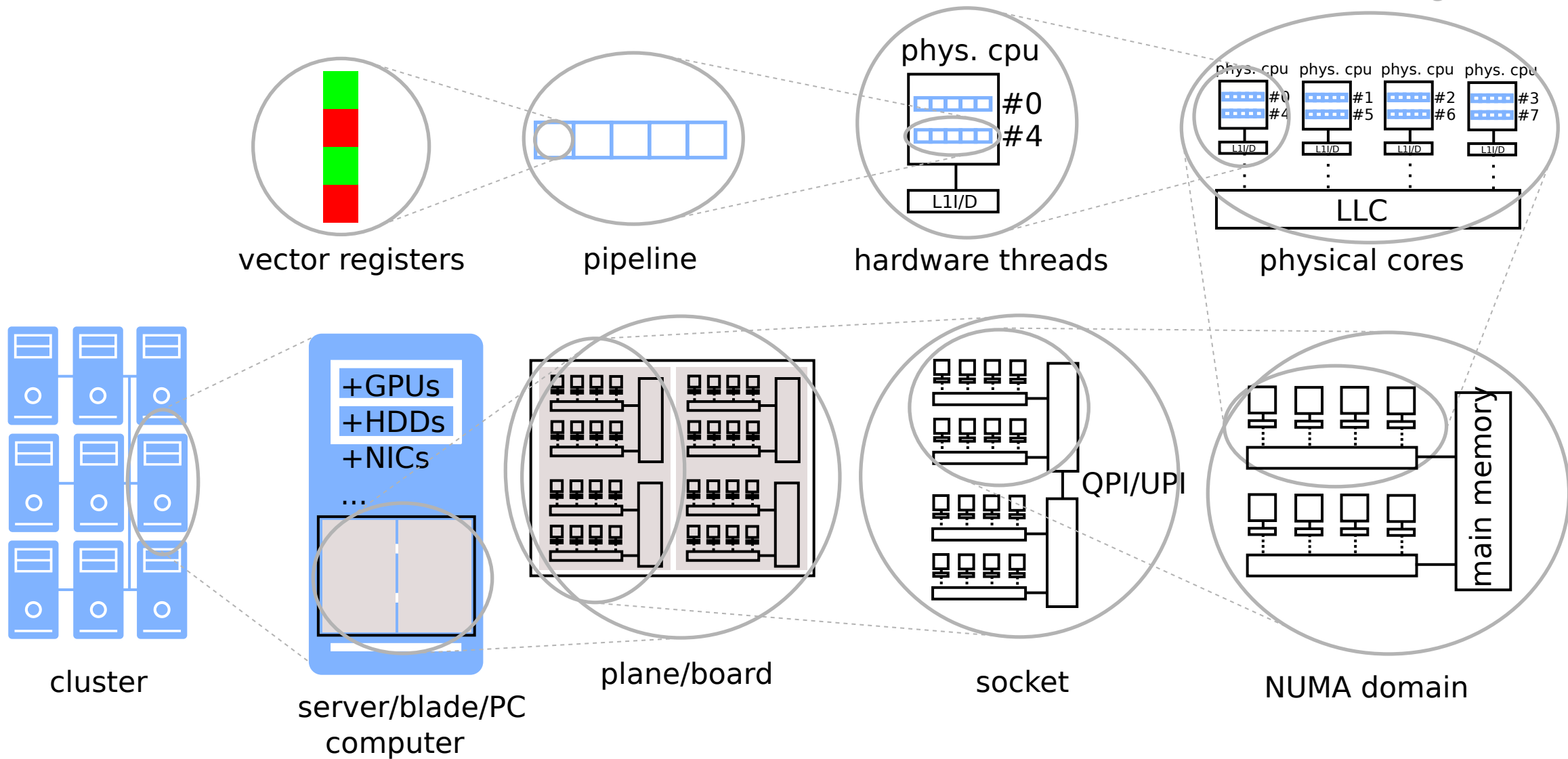
Julia Details

Julia Type System - Numbers



Parallelism

Levels of Parallelism



Parallel Programming Models in Julia



SIMD/Vectorization

Threads

Tasks

MPI